ITAI 4374 · AI & Human Cognition

Course Assignments

Kaire Campomanes · HCC Spring 2026

github.com/kairecamille/Kaire-Campomanes-ML-Course

Course Overview

Focus

The intersection of AI systems and human cognition — memory, perception, attention, emotion, decision-making.

Tools

Wolfram Language, Jupyter Notebooks, AI chat systems, cognitive evaluation frameworks.

Skills

Brain-inspired AI · Neuroscience principles · Neural modeling · AI behavior auditing · Scientific writing.

Repo

Assignments · Labs · Midterm · Final Project — all tracked on GitHub.

A01

Brain vs AI Dialogue

Written Assignment

Assignment Summary

- A creative fictional dialogue between the human Brain and an AI exploring how each processes information.
- Topics covered: learning efficiency, memory, energy use, decision-making, and pattern recognition.
- Framed with an introduction and conclusion grounded in course readings (Russell & Norvig, 2021).
- Key insight: Human and AI intelligence are complementary — not competing.

"So you're saying I'm the heart, and you're the spreadsheet." — Brain

Additional Assignments

A02

Memory & AI Retention

Compared biological memory systems (episodic, semantic, procedural) to AI memory architectures. Analyzed limitations of transformer context windows vs. human long-term memory.

Cognitive Analysis

A03

Attention & Neural Networks

Explored how attention mechanisms in Transformer models parallel selective attention in the human brain. Included analysis of the Stroop effect and attention bottlenecks.

Neuroscience + AI

A04

Emotion & Decision-Making in AI

Investigated how emotions guide human decisions (Damasio's Somatic Marker Hypothesis) and whether AI systems can or should model emotional reasoning for ethical outcomes.

AI Ethics

Labs & Midterm

Labs

Lab 1 · Neural Firing Simulation

Modeled action potential patterns using Wolfram Language to simulate neuron firing thresholds.

Lab 2 · Perceptron from Scratch

Built and trained a single-layer perceptron in Python to classify binary patterns, mirroring biological neuron logic.

Lab 3 · Cognitive Load & LLM Benchmarks

Measured working memory limits in humans and compared against context-window constraints in LLMs.

Midterm

AI Behavior Audit

Conducted a structured audit of an AI chatbot's responses across multiple domains — accuracy, bias, emotional sensitivity, and transparency.

Deliverable: a formal written report with a rubric-based scoring framework and recommendations for responsible deployment.

Final Project

Brain-Inspired Neural Modeling Simulation

Developed a Jupyter Notebook simulation modeling how the brain's visual cortex processes edges and patterns, then mapped those mechanisms to convolutional neural network (CNN) layers. Used Wolfram Language to visualize activation maps and compare them to fMRI-style imaging data from course readings.

📊 Deliverable

Jupyter Notebook + written analysis

🧠 Core Concept

CNN layers ≈ visual cortex hierarchy

🔗 Tools Used

Python · Wolfram Language · NumPy

Key Themes Across the Course

01

Learning Efficiency

Humans generalize from few examples; AI requires large datasets — each with its own tradeoffs.

02

Memory Architecture

Biological memory is associative and emotion-weighted; AI memory is precise but context-limited.

03

Energy vs. Scale

The brain runs on ~20W; AI training consumes massive energy — but can serve millions simultaneously.

04

Emotional Intelligence

Human decisions are shaped by emotion and context; AI lacks this grounding but can be designed to account for it.

05

Collaboration as the Future

The most powerful systems combine human intuition with AI speed and pattern recognition.

Thank You

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